Madhan P

System Software Engineer | Aspiring Linux Kernel Developer

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Profile Summary

System software engineer focused on building software close to the hardware, with a strong foundation in C, Assembly, Linux internals, and operating systems, along with C++ and Python. Hands-on with system calls, file I/O, and low-level debugging, backed by a cybersecurity internship. Aiming to become a Linux kernel developer, actively deepening knowledge of kernel internals, memory management, process scheduling, and device drivers through a build, break, understand, and rebuild approach to learning.

Education

Kongu Engineering College, Erode — CGPA: 9.11 — 2023 – Present
JKKN Matriculation School — Grade 12: 76.5% — 2022 – 2023

Technical Skills

Languages: C, C++, Python, Java, Assembly
Tools: GCC, GDB, Git, NASM
Cloud: AWS – EC2, S3, RDS, Lambda, IoT Core
Security: Vulnerability & Threat Analysis, Security Audits
Domains: System Software, Operating Systems, Cybersecurity
Systems: Linux Internals, OS, System Programming
Scripting: Shell Scripting, Terraform
Identity & Access: IAM, PAM
Practices: Secure Software Development

Experience

- **Hackers InfoTech –** *Cybersecurity Intern* — May 2026 – Sep 2026
  - Performed vulnerability assessment, threat analysis, and security audits as part of the cybersecurity team.
  - Supported incident response and prepared security report documentation for live client projects.

Projects

- **Linux Command Utilities – Re-implementation in C** [GitHub] — *C, GCC, POSIX System Calls*
  - Rebuilt 12 core Linux utilities from scratch (ls, cat, cp, mv, rm, grep, head, tail, mkdir, rmdir, pwd, echo) using open, read, write, unlink, rename, opendir/readdir, and getcwd system calls and file descriptors.
- **AsmVerse – x86/x86_64 Assembly Programming** [GitHub] — *NASM, GDB, Linux*
  - Wrote low-level programs covering registers, memory addressing, stack usage, Linux system calls, file I/O, macros, and C inline-assembly integration, debugged with GDB.
- **SecureWiping – NIST-Compliant Data Wiping & IT Asset Recycling** [GitHub] — *TypeScript, Python, Blockchain*
  - Built a cross-platform (Windows, Linux, Android) secure data wiping solution implementing NIST SP 800-88 and DoD 5220.22-M algorithms to permanently destroy sensitive data before devices are reused, resold, or decommissioned.
  - Designed a master–slave LAN architecture to trigger wipes on connected devices from one system, with blockchain-backed tamper-proof wipe certificates, offline bootable ISO/USB mode, and remote brick mode.
- **Integrated Secure Data Erasure & Advanced File Recovery Tool** — *Digital Forensics, Data Sanitization*
  - Designed and developed a tool for digital forensics and data sanitization that combines secure data wiping with advanced file recovery capabilities.
- **RansomScanner – AI-Powered Ransomware & Phishing Detection** [GitHub] *Python, TypeScript, Gemini, LiveKit*
  - Engineered real-time ransomware detection by monitoring mass file modification and encryption patterns, with Google Gemini phishing-risk scoring for URLs and emails, and LiveKit for instant alerts.

Honors and Awards

- **1st Prize** – SIH 2026 Internal Hackathon (Secure Data Erasure & File Recovery Tool)
- **1st Prize** – SIH 2025 Internal Hackathon (SecureWiping)
- **3rd Prize** – BLS360 Identity AI 2026+ Global Hackathon (IAM & PAM)
- **3rd Prize** – Hackwave 2024 (AI-based Phishing Detection)
- **3rd Prize** – Techno Cultural Fest (Crypto Analysis & Ransomware Intelligence Mapping)

Leadership

- **Coding Club Secretary:** Coordinating technical activities and supporting peer learning initiatives in the department.